

Campus Parking and Exit Carpool Coordination

WORKSHOP NO. 4

Project Members:

- Julián David Muñoz Revelo — 20251020042
- Nicolás Acero Ladino — 20251020044
- Santiago Alexander García Bermeo — 20242020268
- Miguel Mateo Guillén Guzmán — 20251020080

Institution: Universidad Distrital Francisco José de Caldas

Computer Engineering Program, School of Engineering

Course: Systems Analysis & Design

Instructor: Eng. Carlos Andrés Sierra.

1. Executive Summary

This report presents the simulation and validation phase of the Campus Parking and Exit Carpool Coordination system proposed for the Engineering Faculty of Universidad Distrital Francisco José de Caldas. Building on the empirical diagnosis of Workshop 1, the conceptual architecture of Workshop 2, and the socio-technical refinement, risk framework, and implementation strategy of Workshop 3, this workshop evaluates whether the proposed system behaves consistently under realistic operating conditions before any pilot deployment is attempted.

Rather than treating the project as a software artifact in isolation, this workshop validates the system as a broader socio-technical coordination arrangement involving users, guards, institutional rules, physical parking constraints, trust conditions, and operational exceptions. In this sense, the simulations are not only used to test performance, but also to examine whether the design decisions made in previous workshops remain coherent when exposed to demand variability, cancellation behavior, synchronization failures, and adoption dynamics.

Two complementary simulation approaches were developed. The first is a Process-Oriented Simulation based on discrete operational events such as vehicle arrival, parking demand, overflow generation, ride-matching requests, cancellations, and synchronization failures. The second is a Behavior-Oriented Simulation that models trust, adoption, reliance on WhatsApp, parking saturation, and reinforcing or degrading feedback loops over time. Together, these models allow the project to be validated from both an operational and a behavioral perspective.

The results confirm that the proposed design is viable only if certain critical conditions are protected. The 110-space basement capacity is structurally insufficient without coordinated

parking allocation, accountability-based reservation management, and carpool support.

Institutional incentives are necessary to move participation in the coordination system beyond the stagnant baseline. Infrastructure reliability has a decisive role because desynchronization, cancellations, and repeated no-show behavior interact to produce trust collapse.

For these reasons, Workshop 4 does not simply confirm previous decisions; it also refines them. The validation process supports the general direction of the design developed in Workshop 3, but also shows that pilot deployment should include a rainy-day contingency protocol, explicit trust-recovery mechanisms, accountability and participation monitoring mechanisms, close monitoring of synchronization reliability, and further review of the participation structure needed to break the current adoption ceiling.

2. Integration and Continuity of the Workshop

2.1 Simulation as Continuation of Previous Workshops

Workshop 4 is the computational validation stage of a project that evolved progressively across the previous workshops. Its purpose is to transform prior findings into simulation assumptions and test whether the conceptual system remains coherent under realistic operational and behavioral conditions. In this way, the workshop closes the loop between system analysis, conceptual design, risk management, and empirical validation.

Workshop 1 provided the empirical basis for simulation calibration. Field observations and survey results showed recurrent early-morning saturation, approximately 90 overflow motorcycles on surrounding streets, and widespread use of informal coordination channels such as WhatsApp. These observations directly informed demand ranges, overflow thresholds, initial

platform adoption, and the representation of informal coordination as a competing behavioral state.

Workshop 2 contributed the initial architecture of parking reservation, status management, ride matching, and operational supervision. Although that version still leaned more heavily toward platform structure, it already defined the key flows that are now simulated computationally: parking requests, availability allocation, exit-time matching, cancellations, and coordination outcomes.

Workshop 3 reframed the project as a socio-technical coordination system rather than as a web application. It also introduced explicit risks such as low adoption, parking-state desynchronization, misuse, no-show behavior, institutional misalignment, and trust erosion. Those elements are essential in Workshop 4 because the simulations are designed precisely to test how these risks behave under baseline, optimization, and failure conditions.

2.2 Continuity Table

Previous Workshop	Key Finding or Decision	Simulation Translation	Role in Workshop 4
Workshop 1	Recurrent early arrival peak and overflow of about 90 motorcycles	Day-of-week demand ranges, 110-space basement limit, overflow cap	Process-Oriented Simulation
Workshop 1	Informal coordination through WhatsApp is already dominant	Initial WhatsApp usage and competing adoption state	Behavior-Oriented Simulation
Workshop 2	Parking allocation and exit carpool coordination were	Match requests, reservation flow, cancellations,	Process-Oriented Simulation

	defined as central operational components	synchronization events	
Workshop 3	Risks of low adoption, trust erosion, and desynchronization were identified	Adoption bonus/penalty, trust updates, sync failure rate, cancellation rate	Behavior and Process Simulations
Workshop 3	The platform was redefined as one enabling component inside a broader operational system	Validation focuses on system behavior, not only application performance	Entire workshop logic

This continuity is important because it shows that the simulations are not detached technical exercises. They are formal validation mechanisms built directly on the evidence, architecture, accountability framework, and risk logic developed throughout the project.

3. Development of the Simulation Model

3.1 Simulation Paradigm Selection

Two simulation paradigms were selected because the project involves both discrete operational sequences and collective behavioral dynamics. A single model would not capture both dimensions with enough clarity, especially given that the system combines parking-state transitions, user decisions, trust variation, institutional incentives, and failure cascades.

The first paradigm is Discrete-Event Simulation, used for the Process-Oriented model. This choice is appropriate because the parking and coordination system changes state at identifiable events: vehicle arrival, reservation request, allocation decision, overflow generation,

cancellation, no-show event, manual intervention, synchronization issue, and successful or failed ride matching.

The second paradigm is Agent-Based Modeling, used for the Behavior-Oriented model. This is appropriate because adoption and trust do not evolve only from aggregate totals, but from feedback between individual behavior, system reliability, perceived usefulness, and institutional incentives. The behavior model simulates 500 students over 60 academic days, allowing macro-level dynamics to emerge from repeated interactions between users and operating conditions.

3.2 Process-Oriented Simulation

The Process-Oriented Simulation models the operational sequence of campus parking and exit coordination over a seven-day cycle. It includes day-specific motorcycle demand, a fixed parking capacity of 110 spaces, possible rain events, overflow generation, carpool request generation, cancellation behavior, no-show events, accountability conditions, manual intervention events, synchronization failures, and final matching outcomes.

This model reflects the operational logic defined in previous workshops. Parking demand is generated according to day-of-week ranges, overflow is calculated after comparing demand with basement capacity, and carpooling activity is simulated through daily request ranges and probabilistic matching. By including cancellation, no-show behavior, accountability issues, manual intervention requirements, and synchronization problems, the model also tests how non-ideal conditions alter performance beyond idealized flow diagrams.

For methodological clarity, the current Process-Oriented Simulation represents the core operational flow through demand generation, overflow calculation, carpool request generation,

matching attempts, cancellation effects, and synchronization issues. Elements such as manual intervention, accountability escalation, and no-show behavior are conceptually recognized in the broader system design, but in the present simulation version they are treated as part of the operational interpretation of failures rather than as fully independent computational entities. This distinction is important to preserve consistency between the conceptual scope of the project and the current implementation level of the simulation model.

3.3 Behavior-Oriented Simulation

The Behavior-Oriented Simulation models the social sustainability of the system over time. It includes participation in the coordination system, trust, WhatsApp usage, parking saturation, driver participation ratio, accountability conditions, and the effect of repeated success, misuse, no-show behavior, or failure on future engagement. . This model is essential because the conceptual system designed in Workshop 3 depends not only on operational correctness, but also on whether users continue participating instead of reverting to informal coordination.

The simulation begins with 500 students, an initial adoption level of 30, an initial trust score of 50, and a driver ratio of 15 percent among active users. At each simulated day, parking saturation is calculated, carpool matches are executed, and feedback loops modify participation and trust depending on success, cancellation, no-show behavior, misuse events, manual interventions, or synchronization failure. In this way, the model tests whether the system can remain behaviorally viable beyond its initial deployment conditions.

In its current implementation, the Behavior-Oriented Simulation operationalizes trust, adoption, WhatsApp dependence, parking saturation, driver availability, successful matches, cancellations, and synchronization failures as the main behavioral state variables. Accountability,

no-show behavior, and misuse are treated as design-relevant behavioral conditions discussed in the analysis, but they are not yet parameterized as separate state-transition mechanisms in the current computational version. This limitation does not invalidate the model, but it defines the current boundary between conceptual interpretation and implemented simulation logic.

4. Simulation Scenarios and Experimental Design

4.1 Scenario Framework

To validate the conceptual system under contrasting conditions, three scenarios were defined: baseline, optimization, and failure. These scenarios correspond to the workshop requirement of testing normal operation, improved configuration, and stress conditions, and they are directly linked to the risk framework introduced in Workshop 3.

The baseline scenario represents standard institutional operation without strong incentives or advanced architectural improvements. The optimization scenario represents the intended operational target, where institutional incentives are active, synchronization is reliable, and users behave with greater consistency. The failure scenario stress-tests the system under compounding disruption, including low trust, frequent cancellations, and major desynchronization problems.

4.2 Scenario Definitions

Scenario	Objective	Key Parameters	Expected System Behavior
S1 Baseline	Evaluate normal operation without strong support mechanisms	Match rate 45%, sync failure 15%, cancellation 10%, no adoption bonus	Moderate overflow, stable but stagnant adoption, fluctuating trust

S2 Optimization	Evaluate improved architecture and sustained institutional support	Match rate 75%, sync failure 3%, cancellation 4%, adoption bonus +10	Lower overflow, stronger matching, high trust, adoption growth
S3 Failure	Evaluate resilience under severe operational degradation	Match rate 30%, sync failure 40%, cancellation 25%, adoption penalty -30	Persistent overflow, trust collapse, adoption decline, return to WhatsApp

These three scenarios are sufficient to evaluate directional consistency and identify which design decisions are robust, which require reinforcement, and which need revision. Their value lies not only in comparing performance, but in exposing the system's sensitivity to behavior and reliability conditions.

4.3 Variables and Parameters

The operational and behavioral boundaries of the simulation are defined through a set of independent and dependent variables. The most important independent variables are day-of-week parking demand, rain probability, match rate, synchronization failure rate, cancellation rate, adoption bonus or penalty, driver ratio, and average vehicle capacity. The main dependent variables are overflow vehicles, successful matches, cancellations, sync failures, trust score, adoption level, parking saturation, and WhatsApp usage.

These variables are not arbitrary. Demand ranges and overflow limits come from Workshop 1 observations, while adoption, trust, incentives, and failure conditions translate the main human and operational risks already formalized in Workshop 3. As a result, the simulation models remain anchored in the project's earlier evidence and design assumptions.

4.4 Model Assumptions and Limitations

The simulation models presented in this workshop are based on a bounded set of assumptions derived from Workshops 1, 2, and 3. Demand values are modeled through day-of-week ranges rather than through full empirical probability distributions, institutional incentives are represented through simplified adoption bonuses or penalties, and trust dynamics are modeled as aggregated behavioral effects rather than as individual cognitive states. These assumptions were selected to preserve conceptual clarity while still allowing the system to be explored under contrasting scenarios.

The models also have important limitations. First, the current computational implementation does not yet represent all conceptual elements of the broader system with the same level of detail. For example, no-show behavior, accountability escalation, misuse, and manual intervention are central in the conceptual design, but not all of them are parameterized as separate computational events in the present scripts. Second, the simulations rely on scenario-based stochastic behavior rather than on a statistically calibrated dataset from long-term institutional logs. Third, the process and behavior models are complementary but still partially decoupled, which means that some feedback effects are interpreted analytically across models rather than dynamically coupled inside a single simulation engine.

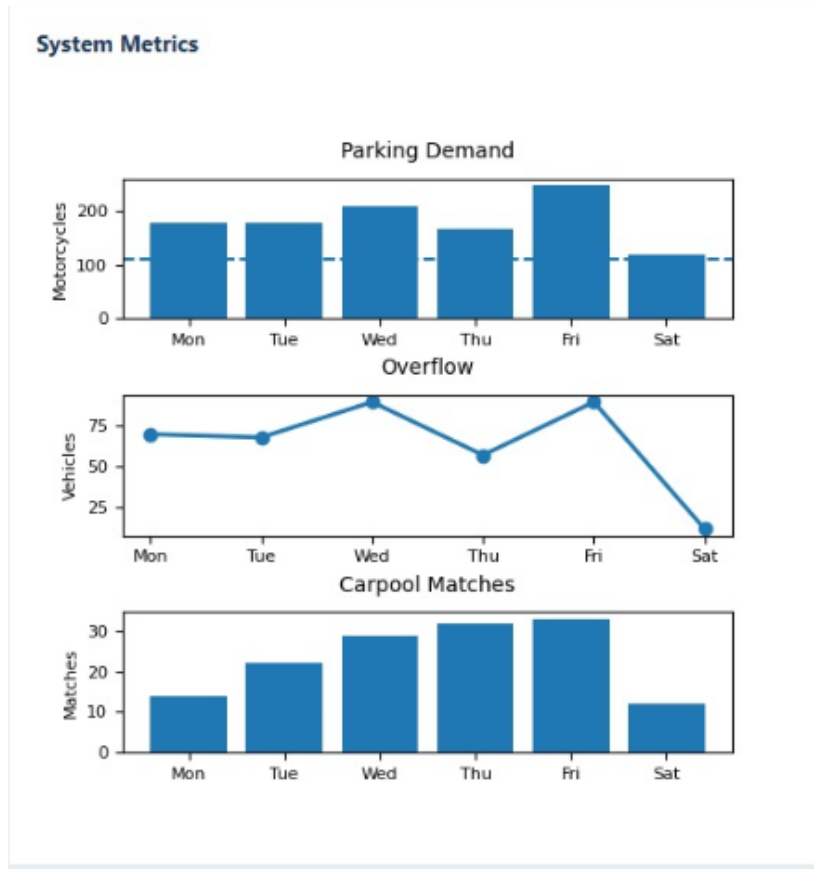
These limitations should not be interpreted as weaknesses of the project definition, but as the natural boundary of the current validation phase. The purpose of Workshop 4 is to provide a meaningful and evidence-based validation of the conceptual system, not to claim that every operational variable has already been simulated at full implementation fidelity.

5. Results and Analysis

5.1 Process-Oriented Simulation Results

The Process-Oriented Simulation was executed over a seven-day operational cycle for the three scenarios. Under the baseline scenario, motorcycle demand fluctuates according to the defined daily ranges, and Friday consistently produces the highest demand. Because the parking basement only supports 110 spaces, overflow occurs regularly on high-demand days, especially when rain amplifies demand by a factor of 1.25. Under these conditions, the system performs moderately, but not enough to eliminate overflow or fully stabilize coordination.

Figure 1 presents system metrics under standard operating conditions. Parking demand exceeds the 110-spot capacity boundary on every weekday, with Friday reaching the highest values while Saturday drops significantly due to lower academic activity. Overflow remains consistently high between Monday and Friday, ranging approximately between 50 and 80 vehicles, and peaks sharply on Friday before dropping to near zero on Saturday. Carpool matches remain low throughout the week, reflecting the 45% match rate limitation, with no day producing more than roughly 30 successful coordinations.

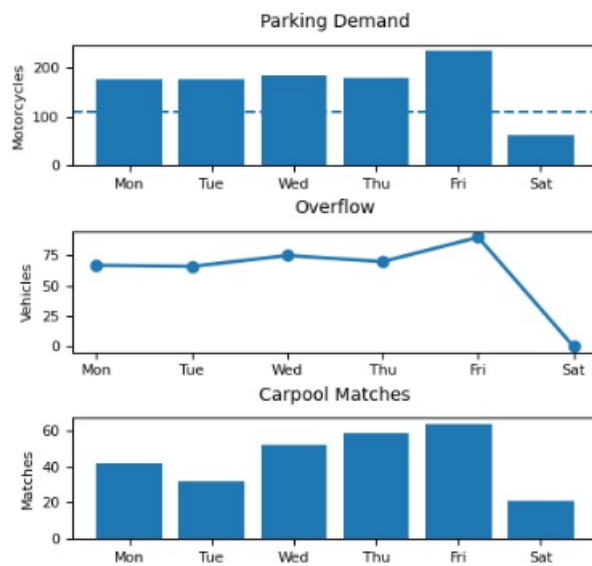


In the optimization scenario, performance improves significantly. The higher match rate and lower failure rates increase successful coordination and reduce operational disruption. As more users are successfully matched, fewer motorcycles arrive individually, which lowers overflow pressure on the physical parking space. This result validates the design assumption that carpool coordination is not a secondary service, but one of the primary mechanisms for reducing saturation.

Figure 2 shows system performance under the optimization configuration. Parking demand follows the same weekly pattern as S1, with all weekdays exceeding the 110-spot capacity boundary and Friday reaching peak demand near 230 motorcycles. Overflow remains

substantial across all weekdays, ranging between 60 and 80 vehicles, with Friday producing the highest spike before dropping to zero on Saturday. However, the key improvement over S1 is visible in the carpool matches panel: the 75% match rate produces significantly higher coordination output across all days, with Friday reaching approximately 60 successful matches nearly double the S1 baseline confirming that match rate directly determines coordination capacity even when physical overflow cannot be fully eliminated.

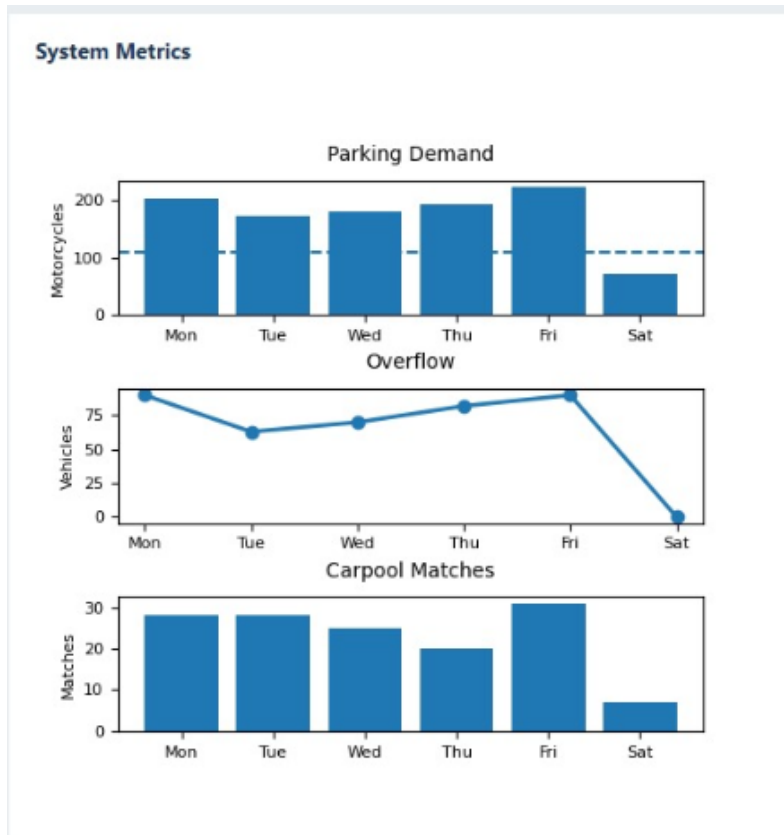
System Metrics



In the failure scenario, the system performs poorly across all metrics. Overflow reaches its maximum boundary repeatedly, synchronization problems become frequent, and cancellations

significantly reduce usable coordination outcomes. This confirms that the operational system is highly sensitive to infrastructure reliability and behavior quality, especially when both deteriorate at the same time.

Figure 3 illustrates system behavior under critical degradation. Parking demand follows the same weekly pattern as S1 and S2, but overflow remains consistently high across all weekdays with no meaningful reduction compared to the baseline. The 30% match rate and 25% cancellation rate produce carpool coordination levels similar to or lower than S1, confirming that infrastructure degradation eliminates the system's primary overflow mitigation mechanism. The similarity between S1 and S3 overflow patterns reveals that without reliable coordination, the system effectively reverts to unmanaged parking conditions regardless of physical demand levels.



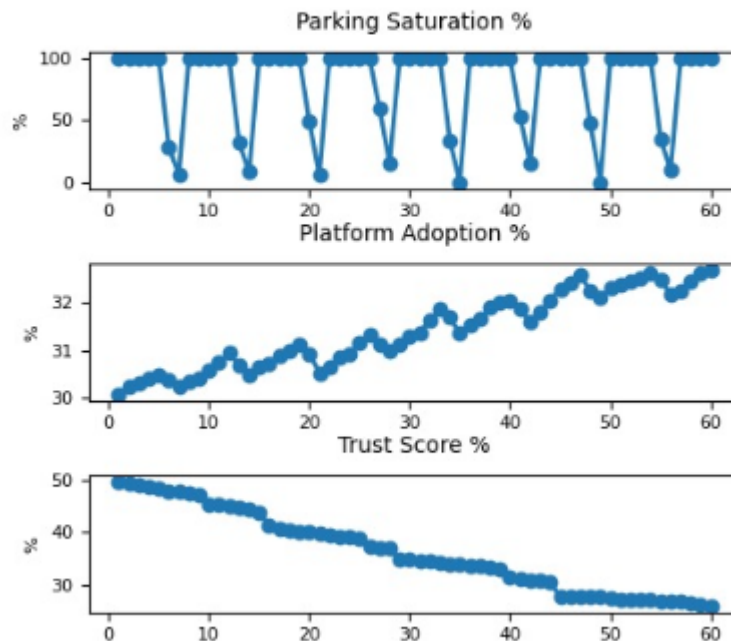
5.2 Behavior-Oriented Simulation Results

The Behavior-Oriented Simulation was executed over 60 operational days. In the baseline scenario, adoption remains close to its initial value of 30, with no strong growth trend. Trust fluctuates around its initial condition because successful matches create some recovery, but cancellations and sync failures repeatedly erode confidence. In this scenario, WhatsApp remains the dominant coordination method, which means the formal system does not become socially central on its own.

Figure 4 tracks behavioral dynamics over 60 operational days without institutional incentives. Parking saturation oscillates sharply between near-zero and 100% in a weekly pattern,

reflecting the day-of-week demand structure established in the Process-Oriented model. Platform adoption starts near 30% and increases gradually throughout the 60-day period. Trust score begins around 45% and degrades steadily, driven by recurring cancellations and sync failures that outpace the recovery produced by successful matches.

System Metrics



In the optimization scenario, sustained incentives and lower failure rates produce a much more favorable equilibrium. Adoption grows toward its upper boundary of 40, trust compounds upward, and WhatsApp dependence declines over time. This result supports the Workshop 3 argument that adoption cannot be assumed as spontaneous; it must be actively supported through institutional incentives and consistent service quality.

Figure 5 shows the compounding effect of sustained institutional incentives over 60 days. Parking saturation maintains the same weekly oscillation pattern as S1, but the overall saturation level is moderated by higher carpool uptake reducing individual motorcycle arrivals. Platform adoption grows consistently from its initial 30% toward its upper boundary of approximately 37-40%, driven by the adoption bonus and reinforced by higher service quality throughout the simulation. Trust score starts near 50% and compounds upward steadily, reaching approximately 55-60% by day 60, as the low cancellation rate and minimal sync failures eliminate the main erosion mechanisms observed in S1 and S3.

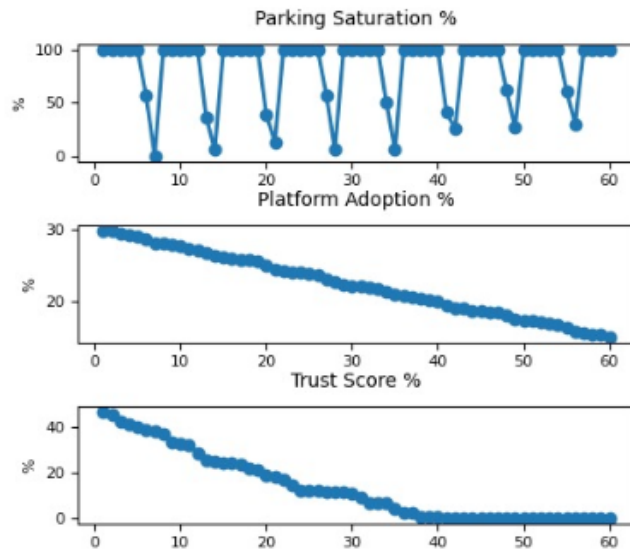
System Metrics



In the failure scenario, adoption falls rapidly toward the minimum boundary of 15, while trust degrades sharply due to the combined effect of cancellations and synchronization failures. The system then converges toward low adoption, low trust, and persistent dependence on informal coordination. This result transforms a qualitative concern from Workshop 3 into a directly observable simulation outcome.

Figure 6 reveals the speed and depth of behavioral collapse under degraded conditions. Parking saturation maintains the weekly oscillation pattern but remains at maximum levels during peak days throughout the entire 60-day period, as coordination failures prevent any reduction in individual motorcycle arrivals. Platform adoption falls rapidly from its initial 30% toward its minimum boundary of approximately 15%, driven by the -30% adoption penalty and compounded by deteriorating service quality that eliminates user incentive to engage with the platform. Trust score degrades sharply from its initial state, approaching near-zero values by the end of the simulation period, eliminating the system's internal recovery mechanism and confirming the compounding cascade described in Section 6.2.

System Metrics



5.3 Comparative Summary

Overflow condition	Moderate, with Friday peaks	Significantly reduced	Maximum, repeatedly at cap
Matching performance	Medium	High	Low
Adoption behavior	Stable near 30	Grows toward 40	Falls toward 15
Trust behavior	Fluctuating near initial state	Compounding upward	Rapid collapse
Informal coordination	Still dominant	Declining	Dominant again

5.4 Model Validation Notes

Both simulations are single-run stochastic models. Because parameters such as daily demand, rain events, match rates, sync failures, and cancellations are probabilistic, repeated independent runs would be required to generate confidence intervals and statistically robust conclusions. This represents a known limitation of the current implementation.

However, three internal consistency checks support the reliability of the results presented in Sections 5.1 and 5.2. First, all dependent variables respond directionally to their corresponding parameter changes: S2 consistently outperforms S1, and S3 consistently produces the worst outcomes across both simulations, with no counterintuitive results observed. Second, the simulation boundaries for adoption and trust function correctly, preventing runaway values while preserving meaningful differentiation between scenarios. Third, the day-of-week demand structure is applied identically across both simulations, ensuring that Process-Oriented and Behavior-Oriented results are directly comparable at the scenario level.

The results are therefore interpreted as behaviorally consistent single-run outputs rather than statistically confirmed distributions. A formal multi-run validation with confidence intervals is recommended before pilot deployment and is identified as a future research direction in Section 8.3.

6. Complexity and Emergent Behavior Analysis

6.1 Parking Saturation as an Adoption Driver

One emergent behavior observed in the simulations is that adoption does not grow smoothly or linearly. Instead, it increases when parking saturation becomes severe, especially above the 90 percent threshold, because users perceive the system as necessary rather than optional. When saturation drops below 60 percent, that pressure weakens and some users reduce their engagement. This means the system's perceived value depends partly on the visibility of the parking problem itself.

This is an important system insight because it suggests that adoption is context-sensitive. If the university improves parking conditions without simultaneously maintaining incentives and coordinated use, the platform may lose engagement over time. The system therefore requires sustained institutional support even when short-term pressure appears to decline.

6.2 Trust Degradation as a Compounding Cascade

A second emergent behavior is the nonlinear collapse of trust under simultaneous failure conditions. In the failure scenario, cancellations and synchronization problems do not simply add their effects; instead, they reinforce each other. As trust falls, matching success decreases. As matching success decreases, the system loses one of its few recovery mechanisms, which makes further trust degradation even more likely.

This finding has direct implications for design. It means that infrastructure reliability and cancellation management should not be treated as separate mitigation targets. They form a joint risk that can destabilize the entire socio-technical arrangement if not addressed together.

6.3 Rain Events as a Demand Amplifier

The simulations also reveal that rain should not be modeled only as a minor environmental disturbance. Because rain increases daily demand by a factor of 1.25, its interaction with already high Friday demand creates a compounded overflow condition. On rainy Fridays, effective demand can rise well beyond the normal saturation pattern and consistently exceed the parking system's manageable range.

This is important because neither "Friday peak" nor "rain event" alone fully represents the actual risk window. Their interaction creates a high-risk operational condition that justifies a dedicated protocol, including earlier notification, overflow redirection, and stronger promotion of coordinated travel on those days.

6.4 Adoption Boundary Behavior

The behavior model enforces adoption boundaries between 15 and 40. Under optimization conditions, adoption reaches the upper boundary but does not exceed it, which reveals a structural ceiling rather than a temporary limit. The simulation suggests that current assumptions about driver ratio and average vehicle capacity create a bottleneck that incentives alone cannot overcome.

This finding is valuable because it shows that the system cannot rely on communication campaigns alone for long-term growth. If the university wants the coordination logic to scale further, it may need to revise participation rules, driver incentives, or practical occupancy conditions.

7. Design Validation

7.1 Validation of Previous Design Decisions

The main purpose of Workshop 4 is to determine whether the key design decisions from Workshops 2 and 3 remain valid under simulation. The results show that most of the core direction is supported, although some decisions require refinement rather than simple confirmation.

Design Decision	Simulation Finding	Verdict	Implication
110-space basement capacity as operational limit	Overflow remains frequent on high-demand days	Validated	Carpool coordination remains necessary
Institutional incentives as adoption driver	Adoption grows only under active support	Validated	Incentives must be sustained
Matching as overflow mitigation	Higher match rate reduces overflow	Validated	Matching efficiency is a central design priority
Reliability as trust condition	Sync failures trigger trust collapse when compounded with cancellations	Validated	Reliability is the highest implementation priority
Adoption becoming self-sustaining above current levels	Ceiling appears at 40 under current constraints	Partially challenged	Driver ratio and capacity need review
Rain treated as routine variability	Rain + Friday peak creates special overload condition	Challenged	A specific rainy-Friday protocol is required

7.2 Model Validation Approach

Because both models are stochastic and scenario-based, validity was assessed through behavioral consistency rather than through formal statistical confidence intervals. The main

validation criteria were directional consistency across scenarios, proper activation of the feedback mechanisms, respect for hard boundaries on adoption and trust, and cross-model coherence between process results and behavior results.

The simulations met these criteria. The optimization scenario outperformed the baseline across all major outcomes, and the failure scenario consistently produced the weakest results. The process and behavior models also remained coherent with each other, since both rely on the same general scenario structure and the same problem logic grounded in previous workshops.

7.3 Validation Criteria and Reliability Boundaries

Model validity was assessed through four complementary criteria. First, directional consistency was verified by confirming that the optimization scenario systematically improved major outcomes relative to the baseline, while the failure scenario systematically degraded them. Second, boundary consistency was checked by verifying that adoption and trust remained within their predefined limits throughout the simulations. Third, cross-model coherence was evaluated by confirming that operational deterioration in the Process-Oriented Simulation corresponded with declining trust and adoption trends in the Behavior-Oriented Simulation. Fourth, conceptual traceability was verified by ensuring that all major simulated variables were directly linked to findings, risks, or design decisions established in previous workshops.

Although formal confidence intervals and repeated-run statistical summaries are not yet included in the present report, the current validation remains methodologically useful because the models behave consistently with their scenario logic and with the empirical problem framing established in Workshop 1. A future refinement should incorporate repeated-run experimentation

and descriptive statistical summaries to strengthen reliability assessment and reduce sensitivity to single-run stochastic variation.

7.4 Parameter Consistency Between Report and Implementation

Parameter	Reported in Document	Current Code Value	Required Action
Behavior baseline sync failure	15%	10%	Unify report or code
Behavior baseline cancellation	10%	12%	Unify report or code
Behavior optimization cancellation	4%	5%	Unify report or code
Behavior adoption bonus	+10	+0.1	Explain scaling or standardize notation
Behavior adoption penalty	-30	-0.3	Explain scaling or standardize notation
Process optimization cancellation	4%	4%	Consistent
Process failure sync failure	40%	40%	Consistent

This consistency review is necessary to ensure that the narrative interpretation of results corresponds directly to the implemented parameter configuration. Any discrepancy between documented scenario settings and executable model values should be resolved before final submission so that the validation claims remain fully traceable and reproducible.

8. Recommendations and Design Improvements

8.1 Immediate Design Adjustments

Based on the simulation evidence, four design improvements should be prioritized before pilot deployment. First, the project should incorporate a **Rainy Friday Response Protocol** that anticipates the compounded effect of peak demand and weather. Second, it should include a **Trust Recovery Mechanism** to interrupt degradation cascades after clusters of cancellations or synchronization failures. Third, the participation structure should be reviewed to understand how the current driver ratio and vehicle capacity assumptions produce the adoption ceiling. Fourth, infrastructure reliability should be treated as the primary non-functional priority of implementation.

These recommendations are not generic improvements. Each one emerges directly from a specific simulation result and therefore strengthens the continuity between analysis, design, risk management, and validation. In that sense, the workshop does not merely describe problems; it translates them into actionable design refinement.

8.2 Incentive Strategy Recommendation

The simulations show that active institutional incentives are necessary from the beginning of deployment. Without them, adoption remains stable but stagnant near the baseline and the platform does not displace informal coordination. For this reason, incentives should not be treated as a temporary promotional measure, but as part of the system's operating conditions during the early and intermediate stages of rollout.

These incentives should be tied to meaningful platform participation rather than simple registration. Otherwise, the system may accumulate nominal users without improving actual

coordination outcomes. This recommendation is consistent with the Workshop 3 perspective that behavior, accountability, and operational legitimacy are central to system viability.

8.3 Future Research and Pilot Monitoring

Future work should refine the model with more detailed arrival data, longer simulation horizons, and broader sensitivity analysis. In addition, once a pilot is implemented, real logs should be compared with simulation predictions in order to recalibrate match rates, overflow probabilities, trust evolution, and adoption dynamics. This would allow the simulations to become living support tools for future iterations of the system.

The pilot monitoring framework should include at least the following indicators: overflow volume during peak periods, unresolved carpool matches, no-show frequency, mismatch rate between reserved and occupied spaces, average intervention time in conflict cases, and trust-related behavior such as repeated misuse or voluntary retention. These indicators also align with the Workshop 3 recommendation to make trust, accountability, and operational control observable rather than assumed.

9. Conclusions

Workshop 4 completes the progression from system diagnosis to conceptual design, risk-aware refinement, and empirical validation. The simulations show that the project has evolved from an initial coordination idea into a more mature socio-technical system whose viability depends on the interaction between physical constraints, institutional rules, user behavior, trust conditions, and operational reliability.

The simulations validate the central role of coordinated parking allocation and exit-time carpooling in reducing overflow and improving parking conditions. They also confirm that low

adoption, cancellations, and desynchronization are not secondary issues, but structural conditions that determine whether the system can remain credible and useful over time. In this sense, the risks identified in Workshop 3 are no longer only conceptual concerns; they are now observable dynamics within the simulated system.

At the same time, the validation process reveals important limits and new conditions. Rainy Fridays create a compounded overload scenario, trust can collapse through reinforcing failure loops, and adoption reaches a structural ceiling under the current participation assumptions. These findings do not weaken the project; instead, they make it more realistic by identifying the exact conditions under which the conceptual design should be strengthened before implementation.

For that reason, the main contribution of Workshop 4 is not only that it supports the previous design. Its deeper value is that it transforms the conceptual model into a testable system, exposes its vulnerable points, and provides evidence-based directions for pilot deployment and future refinement. The project can therefore be considered ready for a controlled pilot phase, provided that reliability, incentives, accountability, and high-risk operating conditions are treated as first-order design concerns rather than secondary adjustments.